

# Probability Theory - Winter term 2025/26

## Personal Solution Set 6

TU Dortmund

### Problem 1

- a) Calculate the characteristic function of  $X \sim \text{Bin}(n, p)$ . Use it to find  $E(X)$  and  $\text{Var}(X)$ .
- b) Prove  $X_n \sim \text{Bin}(n, \lambda/n)$  converges in distribution to  $X \sim \text{Poi}(\lambda)$ .

### Solution

- a) Using the definition of the characteristic function:

$$\varphi_X(t) = E(e^{itX}) = \sum_{k=0}^n e^{itk} \binom{n}{k} p^k (1-p)^{n-k} = \sum_{k=0}^n \binom{n}{k} (pe^{it})^k (1-p)^{n-k}.$$

By the binomial theorem, this simplifies directly to  $\varphi_X(t) = (1 - p + pe^{it})^n$ . Moments using derivatives:  $E(X^k) = i^{-k} \left[ \frac{d^k}{dt^k} \varphi_X(t) \right]_{t=0}$ .

$$E(X) = -i \left[ n(1 - p + pe^{it})^{n-1} i p e^{it} \right]_{t=0} = np.$$

For the second moment:

$$E(X^2) = - \left[ n(n-1)(1 - p + pe^{it})^{n-2} i^2 p^2 e^{2it} + n(1 - p + pe^{it})^{n-1} i^2 p e^{it} \right]_{t=0} = n(n-1)p^2 + np.$$

The variance is  $\text{Var}(X) = E(X^2) - (E(X))^2 = n(n-1)p^2 + np - n^2p^2 = np(1-p)$ .

- b) Substituting  $p = \lambda/n$  into the binomial characteristic function gives:

$$\varphi_{X_n}(t) = \left( 1 - \frac{\lambda}{n} + \frac{\lambda}{n} e^{it} \right)^n = \left( 1 + \frac{\lambda(e^{it} - 1)}{n} \right)^n.$$

Taking the limit as  $n \rightarrow \infty$ :

$$\lim_{n \rightarrow \infty} \varphi_{X_n}(t) = \exp(\lambda(e^{it} - 1)).$$

This is exactly the characteristic function of a Poisson distribution  $\text{Poi}(\lambda)$ . By Lévy's Continuity Theorem, pointwise convergence of characteristic functions implies convergence in distribution.

### Problem 2

Let  $X_1, \dots, X_n$  be i.i.d. random variables with finite mean  $\mu$  and positive variance  $\sigma^2$ . Prove that the estimator  $\hat{\sigma}_n^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2$  is consistent (i.e.  $\hat{\sigma}_n^2 \xrightarrow{P} \sigma^2$ ).

## Solution

Expanding expression for the estimator by adding and subtracting the true mean  $\mu$ :

$$\begin{aligned}\hat{\sigma}_n^2 &= \frac{1}{n} \sum_{i=1}^n ((X_i - \mu) - (\bar{X}_n - \mu))^2 \\ &= \frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2 - 2(\bar{X}_n - \mu) \frac{1}{n} \sum_{i=1}^n (X_i - \mu) + \frac{1}{n} \sum_{i=1}^n (\bar{X}_n - \mu)^2 \\ &= \frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2 - 2(\bar{X}_n - \mu)^2 + (\bar{X}_n - \mu)^2 \\ &= \frac{1}{n} \sum_{i=1}^n (X_i - \mu)^2 - (\bar{X}_n - \mu)^2.\end{aligned}$$

The first term converges in probability to  $\sigma^2$  by the Weak Law of Large Numbers (WLLN). For the second term,  $\bar{X}_n \xrightarrow{p} \mu$  by the WLLN. Applying the Continuous Mapping Theorem (CMT) with the function  $g(x) = x^2$ , get  $(\bar{X}_n - \mu)^2 \xrightarrow{p} 0$ . Since both terms converge in probability, their difference converges to the difference of their limits:  $\sigma^2 - 0 = \sigma^2$ . Thus, the estimator is consistent.

## Problem 3

Let  $P(X_n = 1) = 1 - 1/n^\alpha$  and  $P(X_n = n) = 1/n^\alpha$  for  $n \geq 2$ .

- a) For which  $\alpha > 0$  does  $X_n \xrightarrow{p} 1$ ?
- b) For which  $\alpha > 0$  does  $X_n \xrightarrow{L_p} 1$ ?
- c) Without using (a), find  $\alpha > 0$  for which  $X_n \xrightarrow{d} 1$ .

## Solution

- a) By definition, convergence in probability requires  $\lim P(|X_n - 1| > \epsilon) = 0$ . For any  $0 < \epsilon < n - 1$ :

$$P(|X_n - 1| > \epsilon) = P(X_n = n) = \frac{1}{n^\alpha}.$$

This probability limits to 0 if and only if the exponent  $\alpha > 0$ .

- b) Convergence in  $L_p$  mean requires  $E(|X_n - 1|^p) \rightarrow 0$ . thus calculating the expectation directly:

$$E(|X_n - 1|^p) = |1 - 1|^p \left(1 - \frac{1}{n^\alpha}\right) + |n - 1|^p \left(\frac{1}{n^\alpha}\right) = \frac{(n - 1)^p}{n^\alpha}.$$

For this rational expression to converge to 0 as  $n \rightarrow \infty$ , the degree of the denominator must strictly exceed the degree of the numerator. Therefore,  $\alpha > p$ .

- c) Convergence in distribution requires  $E(f(X_n)) \rightarrow E(f(1)) = f(1)$  for all bounded continuous functions  $f \in C_b(\mathbb{R})$ . Evaluating the expectation:

$$E(f(X_n)) = f(1) \left(1 - \frac{1}{n^\alpha}\right) + f(n) \left(\frac{1}{n^\alpha}\right) = f(1) + \frac{f(n) - f(1)}{n^\alpha}.$$

Since  $f$  is bounded, the difference  $(f(n) - f(1))$  is bounded by some constant  $M$ . For the second term to vanish in the limit (making the whole expression converge to  $f(1)$ ), I need  $\frac{1}{n^\alpha} \rightarrow 0$ , which occurs precisely when  $\alpha > 0$ .